

HR Analyst

EXCEL | POWER BI | TABLEAU | SQL



PRESENTED BY
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Introduction:

The HR Analytics Dashboard is a data-driven project designed to analyze employee information and understand workforce trends. It helps organizations track key HR metrics such as employee performance, attrition, salary distribution, and department insights. The project uses data visualization and analytics tools to convert raw HR data into meaningful insights. These insights support better decision-making and improve employee management strategies.



DATA CLEANING & PREPROCESSING:



CHALLENGES:

- Data contained null values, special characters in the field, currency was not distributed as per the countries.



STEPS TAKEN:

- Removed null values
- Standardized currency & location data
- Categorized price ranges
- Removed special characters in the field



OUTCOME:

- The data is now clean, consistent, and ready for analysis, ensuring accurate insights and reporting



DATA OVERVIEW



DATA SOURCE

HR Analytics Dataset
(Employee Information & Workforce Data)



TOTAL RECORDS

14,999+ Employee Records



KEY VARIABLES

Employee ID, Age, Gender, Department, Job Role, Salary, Experience, Performance Rating, Job Satisfaction, Attrition Status, Work Environment, and Promotion Details



DATA SOURCE TABLES

- Employee Details
- Department Information
- Job Roles
- Salary Records
- Performance & Attrition Data



Observations



- 1 The Sales department has the highest average monthly income among all departments.



- 2 Research & Development shows the lowest average income, while HR falls in the middle range.



- 3 Salary distribution varies across departments, indicating differences in roles and compensation structures.

Average Monthly Income by Department



Summary

The chart highlights that Sales leads in average monthly income, followed by HR, with Research & Development at the lowest. This reflects the varying compensation levels based on departmental roles and responsibilities.

Attrition count



Employee retention is strong, with **761** employees (**79.27%**) remaining in the organization.



Attrition affects **199** employees (**20.73%**) indicating that approximately one in five employees has left the company.

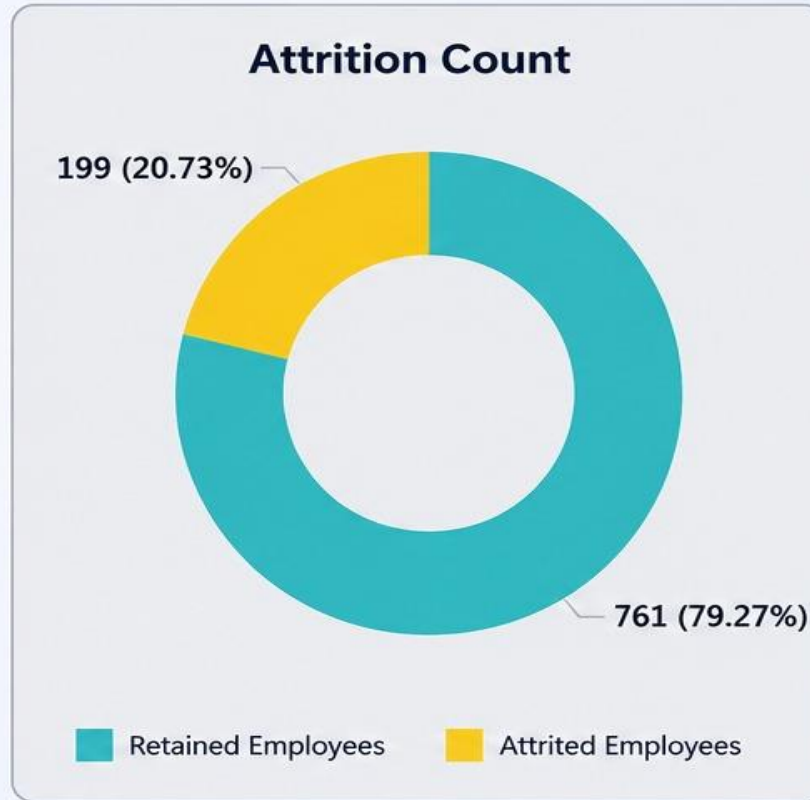


The **low attrition rate** compared to retention suggests relatively good workforce stability and employee retention performance.



Summary

The organization maintains a high retention rate of 79.27%, while the attrition rate stands at 20.73%. This indicates a stable workforce, though reducing attrition further could help improve employee satisfaction and retention.  



Department-wise Employee Count



Research & Development has the highest employee count, indicating it is the largest department in the organization.



Sales has a moderate number of employees, significantly fewer than Research & Development but higher than Human Resources.

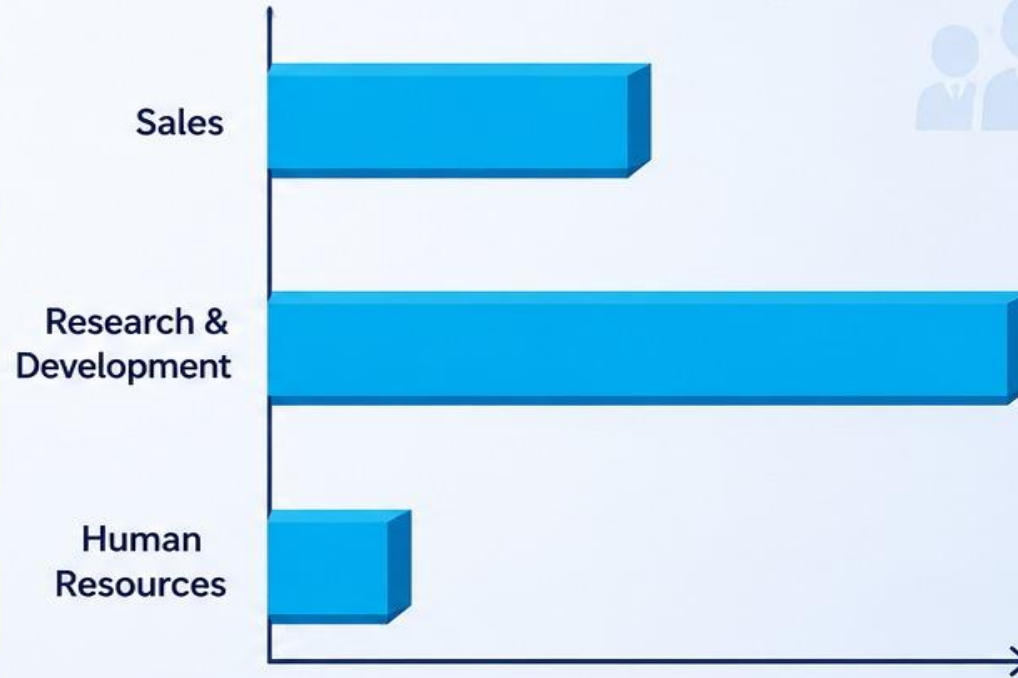


Human Resources has the lowest employee count, suggesting a smaller workforce compared to the other departments.



Summary

The workforce is primarily concentrated in Research & Development, followed by Sales, while Human Resources represents the smallest department. This distribution reflects the organization's stronger focus on operational and business functions.  



Gender Distribution



Male employees dominate the workforce with **884** employees (**60%**), while female employees account for **589** employees (**40%**).



The organization shows a noticeable gender gap, with **295** more male employees than female employees.

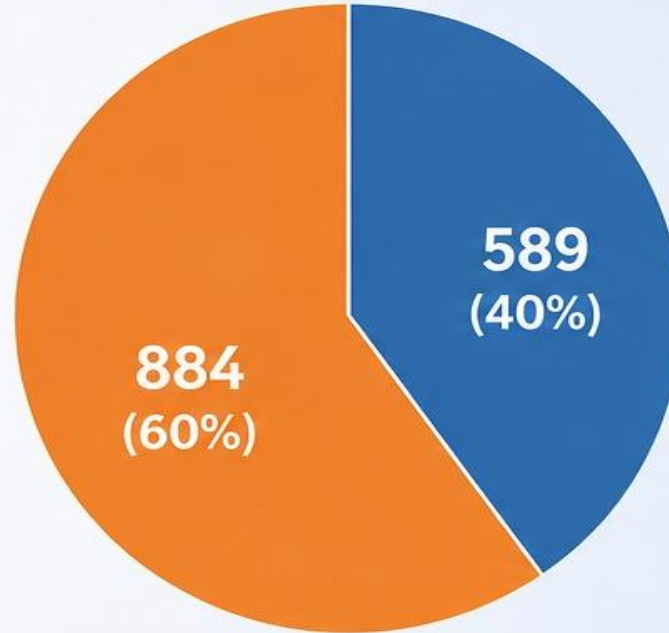


The workforce has representation from both genders, but there is scope to improve **gender diversity and balance**.



Summary

The gender distribution analysis reveals that male employees (884) outnumber female employees (589), accounting for about 60% and 40% of the workforce respectively. Although the organization has representation from both genders, there is a clear male majority, highlighting opportunities to further enhance workforce diversity and inclusion.



Female Employees



Male Employees



KPI List

-  1. Total Employees
-  2. Average Age of Employees
-  3. Max of Monthly Income
-  4. Average Years of Service
-  5. Attrition Count
-  6. Department -wise Employee Count
-  7. Average Monthly income by Department
-  8. Gender Distribution



Excel HR

Analyst Dashboard

Excel HR Analyst Dashboard

Total Employees

1473

Average Age of Employees

36.92

Max of Monthly Income

19999

Average Years of Service

11.28

Department

Human Resources

Research & Development

Sales

Gender

Female

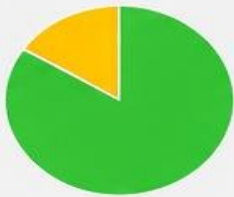
Male

Attrition

No

Yes

Attrition Count



■ No
■ Yes

Department-wise Employee Count



Average Monthly Income by Department



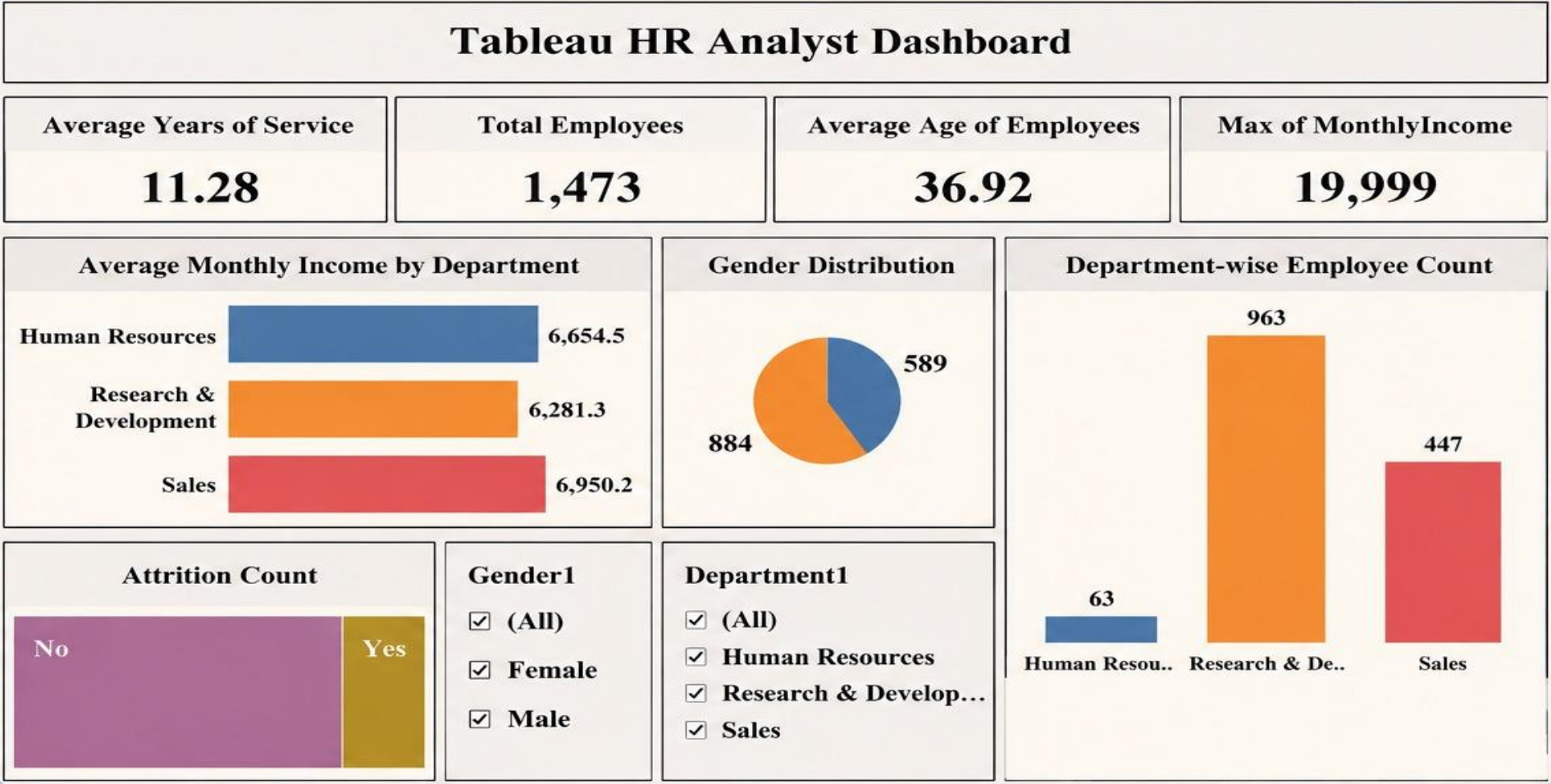
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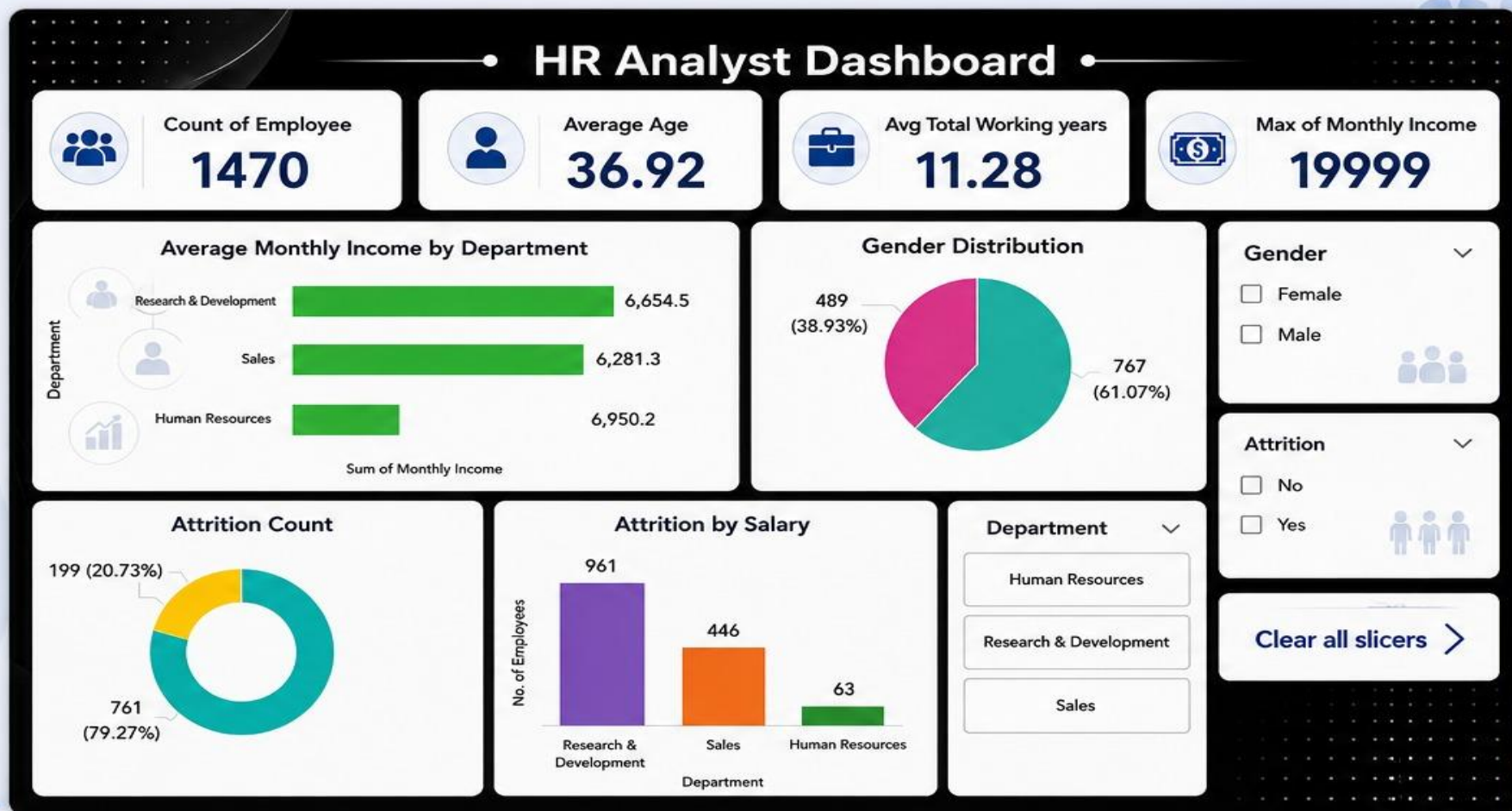
■ Female
■ Male

Tableau HR

Analyst Dashboard



Power BI HR Analyst Dashboard



SQL QUERY FOR EMPLOYEE ANALYSIS

This SQL query calculates key employee metrics including total employees, average age, maximum monthly income, and attrition counts.

SQL QUERY

```
1 • SELECT
2   (SELECT COUNT(*) FROM employee) AS Total_Employees,
3   (SELECT ROUND(AVG(Age),2) FROM employee) AS Average_Age,
4   (SELECT MAX(MonthlyIncome) FROM employee) AS Max_Monthly_Income,
5   SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrition_Yes,
6   SUM(CASE WHEN Attrition = 'No' THEN 1 ELSE 0 END) AS Attrition_No
7 FROM employee;
```

QUERY RESULT

Total_Employees	Average_Age	Max_Monthly_Income	Attrition_Yes	Attrition_No
1473	36.92	19999	237	1236



KEY INSIGHTS

- 1 The query provides a quick summary of key employee KPIs including total headcount, average age, maximum income, and attrition counts.
- 2 The company has **1,473** total employees, giving an overview of the current workforce size.
- 3 The average age of employees is **36.92** years, helping understand the workforce demographic.
- 4 The maximum monthly income is **19,999**, useful for analyzing the salary range within the organization.
- 5 Attrition analysis shows **237** employees have left (Yes), while **1,236** employees are still with the company (No), helping monitor employee retention.

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DETAILED RESULT (IMAGE)

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Analysis	Category	Value
Avg Monthly Income by Department	Research & Development	9462.21
Avg Monthly Income by Department	Sales	6792.29
Avg Monthly Income by Department	Human Resources	6504.71
Avg Monthly Income by Department	Finance	6966.67
Average Age by Department	Research & Development	36.78
Average Age by Department	Sales	35.68
Average Age by Department	Human Resources	36.00
Average Age by Department	Finance	35.50
Average Attrition Age by Attrition	Yes	33.61
Average Attrition Age by Attrition	No	37.56
Attrition Count by Attrition	Yes	237
Attrition Count by Attrition	No	1236



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Conclusion



HR Analytics helps organizations make data-driven decisions by analyzing employee information and trends.



It improves workforce planning, employee performance, and overall business productivity.



Using HR dashboards and reports helps identify problems and create better HR strategies.



Benefits of HR Analyst



HR Analysts help companies understand employee data and improve decision-making.



They support better hiring, retention, performance tracking, and workforce management.



Their insights help reduce costs and increase employee satisfaction.



THANK YOU

